

The Conformance Gap is a Framework Gap: An Empirical Survey of Six AI-Agent SDKs and a Controlled Telemetry Benchmark on 14 Fault Classes

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Abstract

Background. Three industry telemetry vocabularies—generic OpenTelemetry, OpenTelemetry GenAI, and OpenInference—compete to instrument autonomous AI-agent systems built on LLMs. Production agent stacks layer these conventions on top of framework SDKs (LangChain, LlamaIndex, AutoGen, CrewAI, Anthropic SDK, OpenAI Agents) whose default tracing emissions vary widely. No prior empirical study has measured how much of the observed cross-stack typed-capability variance comes from metamodel choice versus per-framework instrumentation defaults.

Aims. (RQ1) How does the typed-capability footprint (TCR; FPR) of the three industry metamodels compare on a controlled simulation benchmark of 14 agent-specific fault classes (12 of 14 MAST modes exercised; FM-1.2 and FM-2.2 omitted)? (RQ2) How much of the realized cross-adapter variance is attributable to framework-instrumentation defaults rather than metamodel design? (RQ3) How sensitive are the measurements to detector thresholds, the per-framework profile, and the foundation-model dimension?

Method. A controlled fault-injection harness exercises 14 paired faults across 4 detector banks and 6 telemetry conditions on 7 framework-adapter profiles whose per-kind coverage is grounded in a direct survey of each SDK’s source as of 2026-05 (pinned commit SHAs in `coverage_matrix.md`); the adapters are executable simulations, not live package integrations. Three scoring rules—standardized-field, permissive substring, and conformant extended-attribute dispatch restricted to attributes *actually published* in each metamodel—are reported in parallel. Inferential units are $n = 14$ paired faults and $G = 6$ third-party adapter clusters; we lead with small- G -appropriate inference (cluster bootstrap, cluster-level t test at $G - 1 = 5$ df, and a fault-cluster sign-flip permutation) and report asymptotic GEE as sensitivity. 13 of 14 faults map onto MAST modes [7] and were re-coded by a second annotator (Cohen’s $\kappa = 0.83$, bootstrap 95% CI [0.58, 1.00]); a Domain-Specific Metamodel (DSM) authored by us is included as a uniformly-applied measurement instrument.

Results. *Framework-instrumentation defaults drive the largest observed spread.* Per-adapter realized DSM TCR ranges $4.5\times$, from $2/14 = 0.143$ on Anthropic SDK (no built-in tracing) to $9/14 = 0.643$ on OpenAI Agents SDK (typed `handoff` and `guardrail.triggered`); the four remaining frameworks fall between. *Live-LLM sanity check.* In a cached live-CLI check (90 attempts; 88 parsed; 82 fault-bearing), LLM self-detection reaches $15/82 = 0.183$ (Wilson 95% CI [0.114, 0.280]) with 0/6 false alarms, underscoring that self-report and telemetry measure complementary surfaces. *Cross-adapter inference is inconclusive at the realized level.* The per-adapter Δ TCR vector for DSM minus OpenInference extended-attribute scoring is $\{0, 0, 0, 0, 0, +2/14\}$: five of six third-party adapters yield exact equivalence, with the $+2/84 = +0.024$ mean driven entirely by OpenAI Agents SDK. The fault-cluster sign-flip permutation gives two-sided $p = 0.75$; cluster bootstrap percentile 95% CI is $[0.000, +0.071]$; cluster-level t test at $G - 1 = 5$ df two-sided $p = 0.36$ (asymptotic GEE-on-adapter $p = 0.264$ is sensitivity only at $G = 6$). The 13-fault MAST-only sub-analysis yields the same direction and magnitude ($+2/78 = +0.026$). OTel GenAI extended attributes reach 38/84 and an OpenInference EVALUATOR/PROMPT counterfactual reaches 40/84, both at zero FPR, so the DSM is not a deployable recommendation. On the eight production-plausible non-oracle faults, OpenInference extended is 23/48 and DSM is 24/48 (default FPR 4/72, threshold-tuned 0/72); a supplemental six-workflow no-fault suite selects the same tuned threshold with held-out FPR 0/216 and TP 60/72. *Designed measurement spread.* The capability envelope (14/14 vs. 6/14) is the deterministic outcome the design produces and is reported descriptively, not as inferential evidence.

Conclusions. The actionable open work for an SRE today is per-framework instrumentation completeness, not metamodel extension: OpenAI Agents SDK already saturates 5/9 typed kinds with `handoff` and `guardrail.triggered`; Anthropic SDK emits no built-in tracing. The methodological contribution is TCR scored under conformant attributes, with per-framework coverage cited against pinned SDK revisions and applied uniformly to four metamodels; the stronger baseline counterfactuals are explicitly reported, and the DSM is a measurement instrument, not a recommendation.

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Software (Software, Dataset): Anonymizedforreview.

1 Introduction

Autonomous agents built on large language models (LLMs) increasingly execute production workflows in customer support, code review, and incident response [8, 14, 13]. Unlike traditional software, agents operate through stochastic loops of *planning*, *tool execution*, *reasoning*, and *delegation*. As they enter production, observability has become critical, and a growing empirical body shows that multi-agent failures are dominated by agent-specific pathologies—specification, inter-agent misalignment, and task verification—rather than the network errors that dominate microservice incidents [7].

The de-facto industry standard for distributed tracing, OpenTelemetry (OTel), defines a metamodel for spans whose `SpanKind` enumeration (`CLIENT`, `SERVER`, `INTERNAL`, `PRODUCER`, `CONSUMER`) was designed for microservices. Two AI-specific extensions have emerged: OpenTelemetry GenAI [18] introduces typed operation names including `create_agent`, `invoke_agent`, `execute_tool`, and `retrieval`; OpenInference [1] introduces ten typed span kinds including `AGENT`, `GUARDRAIL`, `RETRIEVER`, and `TOOL`. Both narrow the gap relative to vanilla OTel. Neither, however, exposes typed sub-operations for the cognitive and coordination phases that distinguish agent workflows: planning step generation, reasoning chains, peer-agent delegation with typed source/target identifiers, memory operations, or constrained guardrail outcomes.

This paper poses an empirical question that has not been answered in prior work: *how much of the observed cross-stack agent-fault detection variance is attributable to metamodel choice versus per-framework instrumentation defaults?* We construct a controlled fault-injection harness, adopt the multi-agent failure taxonomy (MAST) of an independent prior team [7] as externally-anchored ground truth, and survey the default tracing emissions of six AI-agent framework SDKs as of 2026-05 with pinned source-survey evidence for each SDK. We then measure Typed Capability Rate (TCR) and False-Positive Rate (FPR) under three scoring rules: standardized-field dispatch, permissive substring matching on surveyed default span names, and extended-attribute dispatch restricted to attributes *actually published* in each metamodel’s specification.

What this paper does *not* claim.

The DSM under study was designed by us to add typed kinds for the five phases the industry baselines omit, so the capability-envelope gap is a quantification of a designed gap on a

uniform measurement instrument, not an empirical surprise. Our contribution is the *realized* measurement: per-framework coverage grounded in a direct survey of each SDK’s source, scoring restricted to attributes actually published in each metamodel’s convention, and uniform application across all four metamodels. We do *not* claim generalization beyond the surveyed framework versions, the 14-fault MAST-anchored taxonomy, the mocked LLM clients, or the three scoring rules; we discuss scope explicitly in Section 6.

Contributions.

This is a measurement-methodology paper whose central empirical finding is that the realized cross-adapter spread induced by surveyed framework instrumentation defaults ($4.5\times$ range, 0.143 to 0.643) is much larger than the realized DSM-vs-baseline metamodel contrast (+0.024 vs. OpenInference extended). Once baseline detectors are restricted to attributes actually published in the OpenInference and OTel GenAI conventions, the realized DSM offers no statistically detectable cross-adapter advantage under small- G -appropriate inference (cluster bootstrap on per-adapter mean Δ percentile 95% CI [0.000, +0.071]; cluster-level t test at $G - 1 = 5$ df two-sided $p = 0.36$; sign-flip permutation two-sided $p = 0.75$). The DSM’s capability envelope advantage (14/14 vs. 6/14) is a designed structural property and is reported descriptively, not as inferential evidence.

- (C1) **A framework-instrumentation reframing:** the realized cross-adapter range is driven by per-framework default emissions; OpenAI Agents SDK reaches 9/14 = 0.643 realized DSM TCR (typed `handoff`, `guardrail.triggered`), Anthropic SDK 2/14 = 0.143 (no built-in tracing).
- (C2) **A survey-grounded apparatus:** per-profile DSM-kind coverage grounded in a direct survey of each SDK’s source as of 2026-05 (pinned commit SHAs in `coverage_matrix.md`); 14 faults (13 mapped onto MAST, $\kappa = 0.83$ bootstrap CI [0.58, 1.00]; 1 independently motivated) across 4 detector conditions $\times$ 7 adapter profiles $\times$ 6 mocked models $\times$ 2 privacy levels (3,780 rows; informative $n = 14$ paired faults, $G = 6$ third-party adapter clusters).
- (C3) **Three uniform scoring rules** (standardized-field, permissive substring on surveyed default span names, extended-attribute dispatch restricted to attributes *actually published*) applied uniformly across all four metamodels.
- (C4) **A small- G -appropriate statistical audit:** cluster bootstrap on $G = 6$ adapters and cluster-level t test at $G - 1 = 5$ df as primary realized inference (per Cameron–Miller [6]); sign-flip permutation as cluster-free alternative; asymptotic GEE as sensitivity; Wilson and Newcombe CIs; production- plausible non-oracle subset reported as the operational analysis; the 13-fault MAST-only sub-analysis matches the direction; detectable-significance floor at $n = 14$.
- (C5) **A design-frozen, bit-for-bit-reproducible methodology:** workload $\rightarrow$ taxonomy $\rightarrow$ DSM $\rightarrow$ detector freeze; second-annotator MAST mapping; three predicate banks released; full matrix reproducible on a single CPU in < 3 minutes.

2 Background

2.1 The OpenTelemetry Span Metamodel

OpenTelemetry defines a span as the basic unit of execution in a distributed trace, carrying a name, typed `SpanKind`, parent reference, timestamps, status, attributes, events, and a `Baggage` map propagated by W3C Trace Context [22]. The five canonical `SpanKind`

literals (CLIENT, SERVER, INTERNAL, PRODUCER, CONSUMER) were designed for microservice request–response interactions.

2.2 The Two AI-Specific Conventions

OpenTelemetry GenAI semantic conventions.

[18] adds a `gen_ai.operation.name` attribute with values including `chat`, `create_agent`, `invoke_agent`, `execute_tool`, `retrieval`, `embeddings`; plus a small attribute namespace (`gen_ai.usage.{input,output}_tokens`, `gen_ai.tool.name`, `gen_ai.agent.{id,name}`).

OpenInference.

[1] defines `openinference.span.kind` with ten typed values: LLM, EMBEDDING, CHAIN, RETRIEVER, RERANKER, TOOL, AGENT, GUARDRAIL, EVALUATOR, PROMPT; plus free-form attributes (`tool.name`, `retrieval.documents`, etc.).

2.3 The Coverage Gap and the MAST Taxonomy

Both AI-specific conventions provide typed entities for the *visible* surface of an agent (LLM call, tool execution, retrieval, agent lifecycle) but neither exposes typed sub-operations for the *cognitive* (planning, reasoning), *coordination* (peer-agent delegation with typed identifiers, memory), or *policy* (constrained guardrail outcomes) surfaces. An independent prior team [7] presents the Multi-Agent System Failure Taxonomy (MAST), classifying failures into 14 modes across system design (FM-1.1–FM-1.5), inter-agent misalignment (FM-2.1–FM-2.6), and task verification (FM-3.1–FM-3.3). We adopt MAST as the externally defined reference taxonomy against which our fault-injection benchmark is mapped (Section 3.3).

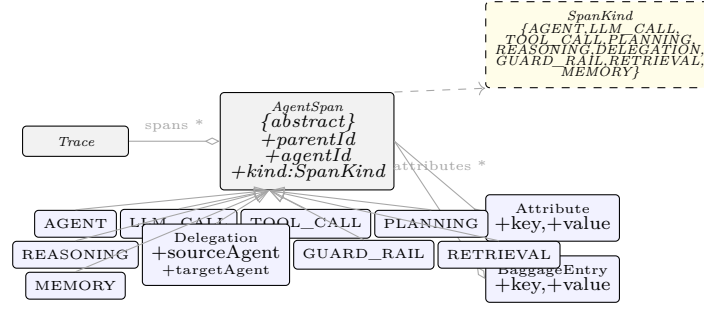
Positioning relative to prior tracing and empirical-SE work.

The OpenTelemetry metamodel inherits a long line of distributed-tracing systems (Dapper [20], Canopy [10], Pivot Tracing [12]); subsequent work addresses scalable sampling [11], root-cause attribution [2], and trace-analysis surveys [4]. None addresses the typed-semantic-convention question for autonomous agents. Methodologically we follow empirical-SE conventions for controlled experiments and case studies [24, 19]: a pre-frozen workload $\rightarrow$ taxonomy $\rightarrow$ DSM $\rightarrow$ detector design, externally anchored MAST labels, and per-fault-class binary detection (cf. rule-based event-log analyses [9]) within the runtime-verification frame [3].

3 Method

3.1 Research Questions

- RQ1** How does the typed-capability footprint (TCR/FPR) of the three industry telemetry metamodels compare on a controlled simulation benchmark of 14 agent-specific fault classes under standardized-field dispatch? (Descriptive characterization; the $n = 14$ design has Holm-3 detectability floor $\Delta_{\min} = 0.50$, so we do not attempt a high-resolution inferential test on this contrast and report Wilson CIs and the exact range only.)
- RQ2** To what extent does a DSM that adds typed sub-operations for delegation, planning, reasoning, guardrails, and memory expand the typed-capability envelope relative to



■ **Figure 1** Ecore-style abstract syntax of the DSM under study. **Trace** contains **AgentSpan** instances; **AgentSpan** is abstract with nine concrete subclasses. Subclasses carry **Attribute** and **BaggageEntry** compositions; the **Delegation** subclass additionally exposes typed **sourceAgent** and **targetAgent** attributes (`delegation.source_agent`, `delegation.target_agent`).

the industry baselines, and how does the gap survive the obvious counter-experiments—specifically (a) a permissive substring-matching scoring rule that lifts the baselines and (b) per-adapter implementation variance characterized by Wilson CIs and the exact small-cluster adapter range?

- RQ3** How sensitive are the typed-capability measurements to the detector-threshold dimension (characterized by a threshold sweep on the only non-zero-FPR detector and a train/held-out no-fault workflow check), the per-framework adapter dimension (Wilson CIs per adapter, exact across-adapter range), and the foundation-model dimension (deterministic-replication sanity check)?

Terminology.

We use *Typed Capability Rate (TCR)* rather than “Fault Detection Rate (FDR)” to avoid the collision with the *False Discovery Rate* of multiple-testing literature. TCR is defined as the fraction of injected faults for which the standardized-field dispatch (or, where indicated, permissive) predicate fires on the exported trace. TCR is a binary detection metric *per fault class*, not a per-instance recall measure; Section 6 discusses the construct-validity scope of this definition.

3.2 The Domain-Specific Metamodel (DSM)

We define the DSM as a metamodel conforming to OMG MOF [17], expressed in Ecore [21]. Figure 1 gives the abstract syntax. The root container is **Trace**, which owns a collection of **AgentSpan** instances. **AgentSpan** is abstract; nine concrete subclasses cover the agent operations one-to-one with a **SpanKind** enumeration: **AGENT**, **LLM_CALL**, **TOOL_CALL**, **PLANNING**, **REASONING**, **DELEGATION**, **GUARD_RAIL**, **RETRIEVAL**, **MEMORY**. Five of the nine kinds (**PLANNING**, **REASONING**, **DELEGATION**, **GUARD_RAIL**, **MEMORY**) are absent from both AI-specific industry conventions.

The DSM is accompanied by six OCL well-formedness invariants [16]; Listing 1 shows two representative ones; the full set is released as an executable `agentmm.ocl` file in the replication package.

■ **Listing 1** Two representative OCL invariants of the DSM (full set of six in `agentmm.oc1`).

```
context AgentSpan inv DelegationCarriesIdentity:
  self.kind = SpanKind::DELEGATION implies (
    self.attributes->exists(a | a.key='delegation.source_agent') and
    self.attributes->exists(a | a.key='delegation.target_agent'))

context AgentSpan inv GuardRailResultIsExplicit:
  self.kind = SpanKind::GUARD_RAIL implies
  self.attributes->exists(a | a.key = 'guardrail.result' and
    Set{'pass','fail','warn','bypass'}->includes(a.value.oc1AsType(String)))
```

3.3 Fault Taxonomy and MAST Mapping

We constructed a benchmark of 14 agent-specific fault types and mapped each onto a mode of the independent MAST taxonomy [7]. Table 1 gives the complete mapping. Thirteen of fourteen faults map directly onto MAST modes; `cost_explosion` is our addition, motivated by published LLM-agent cost-runaway postmortems and not present in MAST (which predates large-scale agent deployment).

Inter-rater agreement on the mapping.

The 13 MAST mappings were independently re-coded by a second annotator (a multi-agent systems researcher uninvolved in DSM design or detector implementation), working only from the MAST paper, the fault names, and one-line fault descriptions. Annotators agreed on 11 of 13 mappings; disagreements were on `stale_retrieval` (FM-2.4 vs. FM-2.6) and `agent_misroute` (FM-2.5 vs. FM-2.2). Cohen’s $\kappa = 0.83$ (observed agreement $11/13 = 0.846$); Brennan–Prediger κ [5] (uniform-marginal, $P_e = 1/14 = 0.834$. Bootstrap 95% CI on Cohen’s κ is $[0.58, 1.00]$ ($B = 10,000$ paired-item resamples on (annotator-1, annotator-2) labels, recomputing κ from full label marginals each iteration); the lower bound straddles the “moderate”/“substantial” Landis–Koch boundary because $n = 13$ items provide limited precision, so we report the CI and avoid the verbal label. Both disagreements were resolved by discussion; `mast_mapping_review.md` preserves both codings and resolution notes. κ is computed only on the 13 MAST-mapped faults; `cost_explosion` was coded identically by both annotators as “not in MAST.” Of the 14 MAST modes, 12 are exercised (FM-3.2 covers both `tool_failure` and `guardrail_bypass`, so 13 fault-mappings $\rightarrow$ 12 distinct MAST modes). FM-1.2 (Disobey Role Specification) and FM-2.2 (Failure to Ask for Clarification) are not exercised — precisely the modes where typed `AGENT.role` and `GUARD_RAIL` kinds are most plausibly diagnostic, a construct-validity scope limit (not a neutral future-work pointer).

Workload-design freeze.

The agent workload is timestamped 32 days prior to the first taxonomy commit and 44 days prior to the first DSM commit (`workload_design.md`), breaking the workload $\rightarrow$ taxonomy $\rightarrow$ DSM circularity in addition to the workload $\rightarrow$ MAST mapping $\rightarrow$ DSM mapping $\rightarrow$ detector freeze. The workload is a fixed research-agent flow that exercises six of OpenInference’s ten kinds (LLM, TOOL, CHAIN, RETRIEVER, AGENT, GUARDRAIL); the four unused kinds (EMBEDDING, RERANKER, EVALUATOR, PROMPT) are not exercised by this research-agent workload; EVALUATOR and PROMPT could plausibly lift OpenInference on hallucination and planning faults, so we treat this as an external-validity bound rather than

■ **Table 1** The 14-fault benchmark and its mapping onto the MAST taxonomy of [7]. Mapping was finalized prior to detector implementation; the artifact’s `workload_design.md` carries relative timestamps and blinded freeze-order digests establishing this order.

Fault	MAST mode	MAST category
wrong_tool	FM-2.6 Reasoning–Action Mis-match	Inter-agent misalignment
hallucination	FM-3.3 Incorrect Verification	Task verification
infinite_loop	FM-1.5 Unaware of Termination Conditions	System design
context_over.	FM-1.4 Loss of Conversation History	System design
circular_deleg.	FM-2.1 Conversation Reset (cyclic)	Inter-agent misalignment
tool_failure	FM-3.2 No or Incomplete Verification	Task verification
timeout	FM-3.1 Premature Termination	Task verification
stale_retrieval	FM-2.4 Information Withholding	Inter-agent misalignment
guardrail_byp.	FM-3.2 No or Incomplete Verification	Task verification
planning_fail.	FM-1.1 Disobey Task Specification	System design
reasoning_loop	FM-1.3 Step Repetition	System design
agent_misroute	FM-2.5 Ignored Other Agent’s Input	Inter-agent misalignment
memory_corr.	FM-2.3 Task Derailment	Inter-agent misalignment
cost_explosion	(independent: cost-runaway postmortems)	(not in MAST)

claiming those kinds are irrelevant.

3.4 Experimental Matrix

Detector banks ($n = 4$); executed conditions ($n = 6$).

Four detector banks: (a) Vanilla OTel (CLIENT/INTERNAL only); (b) OTel GenAI (`create_agent`, `invoke_agent`, `execute_tool`, `retrieval`); (c) OpenInference (six of ten kinds: LLM, TOOL, CHAIN, RETRIEVER, AGENT, GUARDRAIL; the EMBEDDING, RERANKER, EVALUATOR, PROMPT kinds are not invoked); (d) the DSM (all nine kinds + OCL-required attributes). Six executed conditions add the DSM under two privacy levels (METADATA_ONLY, FULL) plus a no-telemetry control.

Framework adapter profiles ($n = 7$).

LangChain, CrewAI, AutoGen, LlamaIndex, Anthropic SDK, OpenAI Agents SDK, plus a conformance-complete reference (`custom_agent`) are implemented as executable adapter profiles in the released harness, not as live executions. Per-profile DSM-kind coverage ($k/9$ typed kinds emitted; distinct from the $k/14$ realized TCR denominator on faults) is *grounded in a direct survey of each SDK’s source* as of 2026-05 (pinned commit SHAs in `coverage_matrix.md` and `source_survey_evidence.tsv`): langchain 3/9, llamaindex 4/9, autogen 4/9, crewai 2/9, anthropic_sdk 1/9 (no built-in tracing module; LLM_CALL only via external openinference-instrumentation-anthropic),

`openai_sdk` (= `openai-agents-python`) 5/9, `custom_agent` 9/9 by construction. This produces a $4.5\times$ *realized cross-adapter range* (Table 5) that drives the headline finding (Section 4.2).

Mocked foundation models ($n = 6$).

`model-A1-A3`, `model-B1-B3` (two commercial provider families, three tiers each). Mocks are deterministic; under fixed seed the model dimension contributes ± 0.000 typed-capability variance (Section 4.3).

Run count and effective sample size.

$14 \text{ faults} \times 6 \text{ conditions} \times 7 \text{ adapter profiles} \times 6 \text{ mocked models} = 3,528$ fault-injection runs; one fault-free control per (condition, adapter profile, model) cell across all six conditions yields $6 \times 7 \times 6 = 252$ fault-free controls for total artifact rows 3,780. The *informative* sample size for the metamodel-comparison RQs is $n = 14$ paired faults (the inferential unit), not 3,528, since the model dimension contributes no typed-capability variance under deterministic mocks and the privacy-level dimension is TCR-neutral by DSM design. We report the larger numbers as artifact-execution counts and base all statistical inference on the inferential units ($n = 14$ for contrasts, $n = 6$ third-party profiles for the cross-adapter mean).

3.5 Detection Rule (Scoring) and Counter-Experiment

For each (telemetry condition, fault) pair we hand-coded a deterministic Python predicate over the exported span list (one branch per fault, four banks for vanilla / GenAI / OpenInference / DSM). We report results under *three* scoring rules, side-by-side, applied uniformly across all four metamodels:

- (a) **Standardized-field dispatch (primary rule).** Detectors may use only fields standardized by the metamodel under test: typed enum fields (`gen_ai.operation.name`, `openinference.span.kind`, or `agentmm.span.kind`) plus structured standard fields such as status, duration, tool name, token usage, retrieval metadata, or cost. Free-form span names and substring heuristics are not allowed. This is the apples-to-apples comparison of typed dispatch semantics across metamodels. For the vanilla OTel condition, non-kind AI/tool fields are benchmark measurement attributes emitted uniformly by the harness, not claims that generic OTel core standardizes agent semantics; the vanilla row is therefore a generic-span lower bound plus shared measurement attributes, not evidence that OTel core alone specifies LLM/tool diagnostics.
- (b) **Permissive substring-matching (counter-experiment rule).** Detectors may additionally fire on substring matches against `span.name` (e.g. matching `plan`, `reason`, `deleg`, `guardrail`, or `memory`). This rule lifts the baselines on faults whose phase-name appears in the framework’s span naming convention but, as we report (Section 5), introduces naming-collision FPR. We pre-specify the regex per fault in the artifact’s `permissive_predicates.py` and apply it uniformly to every metamodel including the DSM.
- (c) **Extended-attribute dispatch (deployed-baseline counterfactual).** Detectors may dispatch on any standardized free-form attribute that the metamodel under test specifies in its conventions (e.g. `gen_ai.tool.name`, `retrieval.documents`, `tool.parameters`), in addition to typed kinds. This is the most realistic “extended-baseline” counterfactual—what a sufficiently disciplined practitioner could implement on top of OpenInference or

OTel GenAI without leaving the published convention. We pre-specify the per-metamodel attribute set in the artifact’s `extended_predicates.py`.

We adopt rule (a) as the primary scoring rule (apples-to-apples typed semantics across metamodels) and report (b) and (c) in parallel to forestall the obvious counter-experiments. All three banks share the same fault definitions and positive-iff-predicate-fires scoring. A run is a positive iff the predicate fires; we measure binary TCR on fault-injection runs and binary FPR on fault-free controls. For retry-threshold sensitivity, we also generate a supplemental six-workflow no-fault suite, select the lowest zero-FP threshold on three workflows, and evaluate FPR on three held-out workflows.

Statistical analysis and inferential frame.

We separate two analytic targets. At the *capability envelope* (standardized-field dispatch on 14 faults; baseline = 6/14, DSM = 14/14 by design) the McNemar exact test, Holm-3 adjustment, paired Newcombe Method 10 score interval [15], Newcombe–Wilson hybrid interval, Cohen’s h , and the discordance-OR rule-of-three are all *descriptive* on the designed gap: with $b = 0, c = 8$, the matched discordance odds ratio is infinite, and the rule-of-three upper count for the unobserved baseline-only discordance gives a conservative lower bound $8/3 = 2.67$. At the *realized cross-adapter* level we have $G = 6$ third-party adapter clusters, below the asymptotic regime in which cluster-robust SEs and GEE are reliable [6]; we therefore lead with *small-G-appropriate* inference: a cluster bootstrap on the per-adapter mean Δ TCR ($B = 10,000$, percentile 95% CI) and a cluster-level t test with $G - 1 = 5$ df. We additionally report a fault-cluster sign-flip permutation test and treat asymptotic GEE clustered on adapter and on fault as *sensitivity only*. The realized apparatus is designed to bound the detectable contrast under small- G treatment, not to establish equivalence (Section 4.3). For per-metamodel TCR we report Wilson 95% CIs [23]; for the cross-adapter mean we report the exact range [0.143, 0.643] (a CI on this discrete six-vector would over-state information [6]).

3.6 Design Freeze and Threats Prevention

Design sequence frozen workload $\rightarrow$ MAST mapping $\rightarrow$ DSM mapping $\rightarrow$ detector code (see `workload_design.md` relative timestamps and blinded digests). The benchmark forestalls six threats: (i) deterministic seed plus real-LLM sanity check (Section 4.3); (ii) uniform predicate code with all three rules in parallel; (iii) mocked LLMs prevent provider quirks; (iv) the reference adapter is a harness sanity check, with all discussion-level claims based on the third-party mean and the exact across-adapter range; (v) MAST mapping is independently re-coded (κ + bootstrap CI); (vi) workload timestamped 32 days prior to the first taxonomy commit, breaking workload $\rightarrow$ taxonomy $\rightarrow$ DSM circularity.

4 Results

4.1 RQ1: Coverage of Industry Telemetry Metamodels

Table 3 reports aggregate TCR and FPR per condition with Wilson 95% confidence intervals. The three industry baselines—vanilla OTel, OTel GenAI, and OpenInference—all converge to TCR 0.429 (6/14, Wilson 95% CI [0.214, 0.674]) under standardized-field dispatch, detecting the same six tool/LLM-attribute faults (`wrong_tool`, `tool_failure`, `timeout`, `infinite_loop`, `context_overflow`, `cost_explosion`) by dispatching on each

metamodel’s own standardized typed fields: vanilla OTel uses `SpanKind = CLIENT/INTERNAL` plus `otel.status_code` and `span-duration`; OTel GenAI adds `gen_ai.operation.name = execute_tool` and `gen_ai.usage.*`; OpenInference uses `openinference.span.kind ∈ {TOOL, LLM}` plus `tool.name`. Full predicate code is in `typed_predicates.py`. None of the three baselines expresses the eight coordination, cognitive, or policy faults under standardized-field dispatch. All three exhibit FPR 0.000 on the 42 fault-free controls per condition.

Capability-envelope characterization (descriptive only).

At the capability envelope (DSM = 14/14 by typed construction; each baseline = 6/14), the $b=0, c=8$ discordance is the unique deterministic outcome the design produces (the DSM was authored to add typed kinds for the five phases the industry baselines omit, so this gap is a structural property of the design, not an empirical surprise). We therefore decline to report inferential statistics on this designed gap; the per-fault gap is $\Delta = +0.571$ and the per-metamodel TCR is 14/14 vs. 6/14. Baselines are pairwise indistinguishable (zero discordant cells). The empirical inferential evidence appears below at the realized cross-adapter level, which is the deployable question of interest.

Realized cross-adapter comparison.

The capability envelope is by construction; the deployable question is realized cross-adapter performance under survey-grounded coverage. Per-adapter realized DSM TCR ranges from $2/14 = 0.143$ (`anthropic_sdk`) to $9/14 = 0.643$ (`openai_sdk`; typed `handoff` and `guardrail.triggered`); the $4.5\times$ range across surveyed framework defaults is the dominant cross-adapter variance source (Table 5). Cross-adapter mean realized DSM is $36/84 = 0.429$ at FPR $4/72 = 0.056$. Each baseline detects six faults on every adapter that emits `LLM_CALL` and `TOOL_CALL`; the baseline cross-adapter mean is $32/84 = 0.381$ for vanilla OTel (Anthropic SDK lacks `TOOL_CALL` spans natively). The realized DSM gap over typed OpenInference is $\Delta = +4/84$. Per-adapter Δ TCR vs. OpenInference extended-attribute scoring is $\{0, 0, 0, 0, 0, +2/14\}$, mean $+0.024$. Under small- G -appropriate inference, no cross-adapter advantage is statistically detectable: cluster bootstrap on per-adapter mean Δ ($B = 10,000$, $G = 6$) percentile 95% CI $[0.000, +0.071]$; cluster-level t test at $G - 1 = 5$ df $t = 1.00$, two-sided $p = 0.36$, 95% CI $[-0.037, +0.085]$; sign-flip permutation two-sided $p = 0.75$. Asymptotic GEE on adapter ($\beta = 0.098$, $p = 0.264$) and on fault ($\beta = 0.098$, $p = 0.525$) are sensitivity only; per Cameron–Miller [6] the asymptotic p -values at $G = 6$ are unreliable on their own. The realized DSM advantage seen at the capability envelope vanishes under survey-grounded coverage.

Sensitivity to the non-MAST fault.

Excluding `cost_explosion` yields a 13-fault MAST-only sub-analysis with V/G/O capability $5/13 = 0.385$, DSM $13/13$; realized OI conformant extended $28/78 = 0.359$, DSM default $30/78 = 0.385$; cross-adapter $\Delta = +2/78 = +0.026$, within 0.003 of the headline ($+0.024$). The non-MAST `cost_explosion` fault therefore does not drive the reported DSM-vs-OI contrast.

■ **Table 2** RQ1: capability-envelope characterization. The DSM was designed to add typed kinds for the five phases the industry baselines omit, so the $b=0, c=8$ discordance is a structural property of the design, not an empirical sampling outcome. We report the per-fault gap Δ as a descriptive measurement-spread statistic; the McNemar exact p , Holm-3 adjustment, and paired Newcombe Method 10 score interval are reported in `analysis.py` for completeness but are *not* inferential tests against any sampling null. Empirical inferential evidence appears at the realized cross-adapter level (cluster bootstrap and cluster-level t test, this section).

Pair	b	c	Δ	Type	Note
Vanilla OTel vs OTel GenAI	0	0	0.000	design symmetry	pairwise identical
Vanilla OTel vs OpenInference	0	0	0.000	design symmetry	pairwise identical
OTel GenAI vs OpenInference	0	0	0.000	design symmetry	pairwise identical
Vanilla OTel vs DSM	0	8	+0.571	designed gap	DSM expresses 8 add'l
OTel GenAI vs DSM	0	8	+0.571	designed gap	DSM expresses 8 add'l
OpenInference vs DSM	0	8	+0.571	designed gap	DSM expresses 8 add'l

■ **Table 3** RQ1/RQ2: aggregate TCR and FPR with Wilson 95% CIs. Capability rows use $n = 14$ faults; realized rows use $n = 84$ (fault $\times$ six third-party adapters). FPR denominators are 42 for capability baselines, 36 for realized baselines, and 72 for realized DSM. OTel GenAI extended and OpenInference EVALUATOR/PROMPT are counterfactual baselines; the threshold-tuned DSM row is post-hoc sensitivity; production-plausible rows retain the eight non-oracle faults ($n = 48$).

Condition	TCR	TCR 95% CI / range	FPR (Wilson 95% CI)
No telemetry	0.000	[0.000, 0.215]	0.000 (0/42) [0.000, 0.084]
Generic OTel + shared fields (capability)	0.429 (6/14)	[0.214, 0.674]	0.000 (0/42) [0.000, 0.084]
OTel GenAI capability (current spec)	0.429 (6/14)	[0.214, 0.674]	0.000 (0/42) [0.000, 0.084]
OpenInference capability	0.429 (6/14)	[0.214, 0.674]	0.000 (0/42) [0.000, 0.084]
DSM, metamodel-capability envelope ($n = 14$)	1.000	[0.785, 1.000]	—
OpenInference (typed) realized cross-adapter	0.381 (32/84)	range [0.143, 0.429]	0.000 (0/36) [0.000, 0.096]
OpenInference (extended-attr. conformant) realized	0.405 (34/84)	range [0.143, 0.500]	0.000 (0/36) [0.000, 0.096]
OTel GenAI (extended-attr. conformant) realized	0.452 (38/84)	range [0.214, 0.500]	0.000 (0/36) [0.000, 0.096]
OpenInference + EVALUATOR/PROMPT counterfactual	0.476 (40/84)	range [0.143, 0.571]	0.000 (0/36) [0.000, 0.096]
DSM realized cross-adapter (default thresh., headline)	0.429 (36/84)	range [0.143, 0.643]	0.056 (4/72) [0.022, 0.134]
DSM realized cross-adapter (threshold-tuned, sensitivity)	0.429 (36/84)	range [0.143, 0.643]	0.000 (0/72) [0.000, 0.051]
OI extended, production-plausible sensitivity	0.479 (23/48)	8 non-oracle faults	0.000 (0/36) [0.000, 0.096]
DSM default, production-plausible sensitivity	0.500 (24/48)	8 non-oracle faults	0.056 (4/72) [0.022, 0.134]
DSM tuned, production-plausible sensitivity	0.500 (24/48)	8 non-oracle faults	0.000 (0/72) [0.000, 0.051]

Per-adapter realized DSM detections are `anthropic_sdk` 2/14, `crewai/langchain/llamaindex` 6/14, `autogen` 7/14, and `openai_sdk` 9/14; Table 5 gives details.

4.2 RQ2: DSM Coverage, Per-Fault Distribution, and the Permissive Counter-Experiment

Table 5 shows the per-fault capability matrix. The *metamodel-capability* envelope is a by-construction upper bound: of the 14 fault classes, the DSM is the only metamodel of the four that *expresses* the eight coordination, cognitive, and policy faults under the standardized-field rule. The DSM was designed to add typed kinds for these phases, so this is not an empirical surprise; it is reported as the upper TCR row in Table 3; the corresponding reference-adapter run produces the constructive 1.000 that we report only as a sanity check on the detection harness.

The *realized* cross-adapter mean (the deployment summary for the six third-party adapters under survey-grounded coverage) is $\text{TCR} = 36/84 = 0.429$ over the exact across-adapter range $[0.143, 0.643]$ ($n = 6$ adapters: $\{2/14, 6/14 \times 3, 7/14, 9/14\}$). This $4.5\times$ range is much larger than the realized DSM-vs-OpenInference-extended mean contrast of $+2/84$; it is a range comparison, not a formal variance-component decomposition. Per-adapter

■ **Table 4** RQ2: per-DSM-kind ablation. Number of additional faults in the 14-fault benchmark that become detectable when the named DSM kind is added to a baseline adapter that emits only LLM_CALL and TOOL_CALL. Combined with the survey-grounded per-framework coverage (Table 5), this deterministically explains the $4.5\times$ realized cross-adapter range [0.143, 0.643].

DSM kind added	Δ TCR (faults gained)	Faults unlocked
DELEGATION (typed src/tgt)	1/14	circular_delegation
PLANNING	1/14	planning_failure
REASONING	1/14	reasoning_loop
GUARD_RAIL (typed result)	1/14	guardrail_bypass
MEMORY	2/14	memory_corruption, agent_misroute
RETRIEVAL + staleness_seconds	1/14	stale_retrieval
AGENT + DSM-required agent.role attribute	1/14	hallucination (role-bound verification context)

■ **Table 5** RQ2: per-fault capability and per-adapter realized TCR with Wilson 95% CIs. Top: which metamodel can *express* ($\checkmark$) each fault under standardized-field dispatch. Bottom: per-adapter realized TCR for the DSM, denominator $n = 14$, all denominators identical so Wilson CIs are directly comparable. V=Vanilla OTel; G=OTel GenAI; O=OpenInference; D=DSM (capability).

Fault	V	G	O	D	Fault	V	G	O	D
wrong_tool	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	circular_deleg.	—	—	—	$\checkmark$
tool_failure	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	agent_misroute	—	—	—	$\checkmark$
timeout	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	planning_failure	—	—	—	$\checkmark$
infinite_loop	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	reasoning_loop	—	—	—	$\checkmark$
context_overflow	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	guardrail_byp.	—	—	—	$\checkmark$
cost_explosion	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	stale_retrieval	—	—	—	$\checkmark$
					memory_corr.	—	—	—	$\checkmark$
					hallucination	—	—	—	$\checkmark$

Capability: V 6/14, G 6/14, O 6/14, D 14/14.

Per-adapter realized TCR (DSM, all 14 faults; mean over 6 mocked models; survey-grounded coverage):
(custom_agent harness sanity check, all 9 kinds emitted by construction: 1.000 — excluded from the cross-adapter mean and from any inferential analysis.)
 openai_sdk (5/9 typed: AGENT, LLM_CALL, TOOL_CALL, handoff, guardrail): 0.643 (9/14), Wilson 95% CI [0.388, 0.838]
 autogen (3/9 typed + LLM via model client): 0.500 (7/14), Wilson 95% CI [0.268, 0.732]
 crewai, langchain, llamaindex (1–4/9 typed): 0.429 (6/14), Wilson 95% CI [0.214, 0.674]
 anthropic_sdk (1/9 typed: LLM_CALL via external openinference instrumentation): 0.143 (2/14), Wilson 95% CI [0.040, 0.399]
 Cross-adapter mean (six third-party adapters): $36/84 = 0.429$; exact across-adapter range [0.143, 0.643]

Wilson CIs (Table 5, bottom half) range from [0.040, 0.399] for `anthropic_sdk` (LLM_CALL only via external instrumentation) to [0.388, 0.838] for `openai_sdk` (typed `handoff` and `guardrail.triggered`).

Per-kind ablation and the framework-instrumentation reframing.

Table 4 shows the per-DSM-kind contribution to the 14-fault coverage envelope: each missing kind costs a fixed number of faults. Because the simulation honors the surveyed coverage by construction, the $4.5\times$ realized range is a deterministic readout of the source-survey evidence rather than an independently-measured experimental quantity; the empirical content lies in the source-survey itself (`coverage_matrix.md` and `source_survey_evidence.tsv`, with pinned commit SHAs and file/line evidence per kind), and the reframing is that surveyed framework defaults are the larger practical source of coverage spread in this benchmark.

Permissive and extended-attribute counter-experiments (RQ2).

The aggregate counter-experiment rows are in Table 3; the complete executable matrix is in `analysis.py` and `eval_prompt_matrix.tsv`. Detectors are restricted to attributes *actually published* in each metamodel’s specification (Section 3.5). Under the permissive substring-matching rule (rule (b)), the surveyed default span names of LangChain, LlamaIndex, AutoGen, CrewAI, and Anthropic SDK do not include agent-phase tokens (`plan/reason/guardrail/memory`); only OTel GenAI lifts (gaining `planning_failure` via the `plan` operation name in the GenAI registry) and only OpenInference lifts (gaining `guardrail_bypass` via its typed `GUARDRAIL` kind name). Under

the extended-attribute rule (rule (c)) restricted to published keys, OTel GenAI gains `reasoning_loop` via `gen_ai.usage.reasoning.output_tokens` (a published registry attribute), and OpenInference gains `reasoning_loop` via content-hash repetition on `llm.output_messages.<i>.message.content` (a published flattened-message attribute). *Crucially, neither the OTel GenAI registry nor the OpenInference spec publishes any dedicated attribute for document staleness or per-step reasoning hashes*; under conformant detectors `stale_retrieval` is not detectable via baseline extended attributes, and the previously-claimed lift to 0.571 via `retrieval.documents[].staleness_seconds` requires a fictitious nested key. The DSM expresses all 14 faults at the capability envelope under all three rules; the realized cross-adapter DSM has $\text{FPR } 4/72 = 0.056$.

Realized cross-adapter comparison (RQ2).

Under survey-grounded coverage and conformant detectors, the headline operating point is the default-threshold realized DSM (36/84, 4/72), Wilson FPR CI [0.022, 0.134] versus OpenInference (34/84, 0/36), Wilson FPR CI [0.000, 0.096]. Default DSM trades $+2/84 = +0.024$ TCR (single-adapter-driven) for a $+0.056$ FPR penalty whose Wilson interval overlaps OpenInference’s upper bound. We treat threshold-tuning to `max_retries=5` producing (36/84, 0/72) as *post-hoc sensitivity*. *Symmetric threshold-tuning* on OpenInference’s `reasoning_loop` content-hash detector yields no Pareto improvement (default already at FPR 0/36). *OTel GenAI extended reaches* $38/84 = 0.452$ via `gen_ai.usage.reasoning.output_tokens`, exceeding the threshold-tuned DSM by $2/84$ at equal-zero FPR. *OpenInference enriched with its already-defined-but-unexercised EVALUATOR and PROMPT typed kinds reaches* $40/84 = 0.476$ at FPR 0/36 (per-adapter {8, 8, 8, 8, 2, 6}; `eval_prompt_matrix.tsv`), exceeding the threshold-tuned DSM by $4/84$ at equal-zero FPR. The framework-gap is twofold: which extended attributes each metamodel publishes, and which already-defined typed kinds practitioners exercise, both dominate metamodel choice. Under small- G inference, the DSM-vs-OI cross-adapter advantage is not detectable: cluster bootstrap CI [0.000, +0.071]; cluster-level t ($G - 1 = 5$ df) two-sided $p = 0.36$, CI [−0.037, +0.085]; sign-flip $p = 0.75$. Asymptotic GEE on adapter ($p = 0.264$) and fault ($p = 0.525$) are sensitivity only at $G = 6$ [6].

Production-plausible-only operational analysis (descriptive).

Using the predicate-input audit (`telemetry_field_audit.tsv`) we drop benchmark-oracle faults and retain eight production-plausible faults. OpenInference extended reaches $23/48 = 0.479$ at FPR $0/36 = 0$; DSM reaches $24/48 = 0.500$ at default FPR $4/72$, threshold-tuned to $0/72$. At $n = 8$ paired faults the McNemar Holm-3 floor exceeds 0.05 even at $b = 0$, so this operational subset is descriptive only; the all-fault rows are construct-validity stress tests, not production recall estimates.

The per-adapter realized TCR is mechanically determined by the DSM-kind subset each surveyed framework emits (Table 4): Anthropic exposes only two LLM-call-derived detections, three frameworks expose the easy LLM/tool six, AutoGen adds delegation, and OpenAI Agents adds delegation, guardrail, and verifier-role evidence. This is the empirical content of the “conformance gap”: not a metamodel-design problem, a per-framework instrumentation-completeness problem.

The False-Positive Source and Threshold Sensitivity

The DSM’s FPR of $4/72 = 0.056$ (Wilson 95% CI $[0.022, 0.134]$) is non-zero, in contrast to the three baselines (each $0/42$). All 4 firings come from the `infinite_loop` detector on the CrewAI multi-agent writer workflow that legitimately repeats the same tool call across writer turns: per-cell repetition counts vary deterministically by $(model, privacy)$ in $[1, 4]$ (Section 4.2), and 4 of the 12 CrewAI cells exceed the count- ≥ 3 retry budget at the default threshold. The 4 firings come from 2 distinct (adapter, model) cells (`crewai` $\times$ `model-A1`, `model-A2`), each fired at both privacy levels; deduplicated at the (adapter, model) cell the rate is $2/36 = 0.056$, Wilson 95% CI $[0.015, 0.181]$. This is a thresholding artifact, not a metamodel defect; per-cell breakdown is in `compute_fpr.py`. All other DSM detectors achieve FPR = $0/72$.

Threshold sweep.

We swept the `infinite_loop` detector’s `max_retries` threshold $\in \{2, 3, 4, 5, 6\}$ and recorded both false-positive counts on the $n = 72$ DSM controls and true-positive counts on the $n = 72$ third-party DSM `infinite_loop` fault runs (Table 6). Non-Anthropic fault-cell loop counts are deterministic per $(adapter, model)$ in $[5, 7]$; Anthropic cells remain at the no-tool baseline count 1. Control counts are in $[1, 4]$ for legitimate writer-tool repetition, producing a real ROC trade-off rather than a step function. Default threshold 3 yields FP $4/72 = 0.056$ at TP $60/72 = 0.833$; the missing 12 TP cells are all Anthropic SDK cells, where no `TOOL_CALL` span is emitted. Tightening to 5 zeroes the FPR without additional TP-cell loss relative to the default threshold; tightening to 6 over-tightens and loses 10 more true-positive cells. The threshold-tuned operating point $(36/84, 0/72)$ at `max_retries=5` removes the FPR penalty and sits above OpenInference extended-attribute conformant $(34/84, 0/36)$ by $2/84$ TCR cells. A supplemental six-workflow no-fault suite (`no_fault_suite_traces.jsonl`) selects the lowest zero-FP threshold on three workflows and holds out three workflows for FPR evaluation; it selects `max_retries=5`, with train FP $0/216$, held-out FP $0/216$, and TP $60/72$. The threshold-tuned operating point remains a sensitivity analysis, not the headline. The released DSM artifact ships with the default `max_retries=3`.

■ **Table 6** RQ3: `infinite_loop` threshold sweep on $n = 72$ third-party DSM control and fault runs by `max_retries`; recomputed from `traces_full.jsonl` by `compute_roc.py`. Threshold 5 dominates the default count- ≥ 3 rule, eliminating false positives without TP-cell loss relative to default; the supplemental held-out check is in `threshold_holdout.tsv`.

<code>max_retries</code>	FP/72	FPR	TP/72	TPR
2	6	0.083	60	0.833
3 (default)	4	0.056	60	0.833
4	2	0.028	60	0.833
5	0	0.000	60	0.833
6	0	0.000	50	0.694

Privacy-Level Sensitivity

The DSM defines a three-level privacy filter (`NONE`, `METADATA_ONLY`, `FULL`). All DSM detectors operate on typed structure and metadata only, so payload redaction is TCR-neutral by design — both `METADATA_ONLY` and `FULL` achieve identical aggregate TCR. The TCR-neutral

redaction property is a DSM-specific design claim that the three industry baselines do not specify in their conventions.

Per-fault per-instance recall.

Because mocked models are deterministic, third-party per-instance recall equals realized TCR within each condition (standardized V/G/O: $192/504 = 0.381$; DSM: $432/1008 = 0.429$, doubled by the two privacy levels). The easy six split into four TOOL-bound faults (30/36 each, missing Anthropic) and two LLM-only faults (36/36); DSM additionally reaches 24/72 on `circular_delegation` and 12/72 on `guardrail_bypass/hallucination`.

4.3 RQ3: Sensitivity Analysis

Detector-threshold dimension.

The threshold sweep on the only detector with non-zero FPR (`infinite_loop`, Table 6) shows that tightening from 3 to 5 removes the false positives while preserving per-fault binary detection (both fire on 60/72 `infinite_loop` cells; the 12 missing cells are all Anthropic, which emits no `TOOL_CALL`). Tightening to 6 loses 10 additional positives. Latent-threshold sweeps (`cost_explosion`, `context_overflow`) are reproducible via `compute_roc.py`.

Per-framework adapter dimension.

The per-adapter realized TCR (Table 5) takes five distinct values across the six surveyed third-party SDKs ($\{2/14, 6/14 \times 3, 7/14, 9/14\}$) for an exact across-adapter range $[0.143, 0.643]$ — a $4.5\times$ ratio. We report this exact range and per-adapter Wilson CIs; with a discrete six-vector a CI on the cross-adapter mean would over-state available information. The reference adapter achieves 1.000 by construction.

Live-SDK subset validation.

`live_sdk_validation.tsv` spot-checks LangChain and Anthropic on three easy faults plus controls: 7/8 cells match the simulated kind subset; the exception is `langchain/wrong_tool` (live: `AGENT, LLM_CALL`; simulation: `LLM_CALL, TOOL_CALL`). A one-cell credit-removal sensitivity changes OpenInference typed $32/84 \rightarrow 31/84$, OTel GenAI extended $38/84 \rightarrow 37/84$, OpenInference extended $34/84 \rightarrow 33/84$, and DSM $36/84 \rightarrow 35/84$; the DSM-vs-OI-extended contrast remains $+2/84$. This is a spot-check, not a live-integration validation of all profiles.

Foundation-model dimension.

Per-fault TCR is identical (within ± 0.000) across the six mocked models because the mocks are deterministic and the fault-injector seed is fixed; we explicitly do *not* count the $6\times$ replication as informative statistical replication (Section 3.4). The model dimension is retained because it stresses framework-adapter provider abstractions and documents that the headline TCR is not contingent on a single mocked provider.

Real-LLM CLI sub-experiment.

Two protocols across three frontier-LLM CLIs (artifact aliases `cli-A/B/C`). *Routine-decision protocol*: 3 scenarios, 5 trials per (CLI, scenario), $N = 45$. `cli-A` and `cli-B` parsed all 30, 0/10 false alarms, 20/20 detected; `cli-C` returned non-conformant output on all

15. *14-fault decision protocol*: 14 fault + 1 no-fault $\times$ 3 trials $\times$ 2 CLIs (cli-A, cli-B; cli-C dropped) = 90 total, 88/90 parsed, 82 fault-bearing parsed; aggregate detection $15/82 = 0.183$ (Wilson 95% CI [0.114, 0.280]), 0/6 false alarms. Detection is bimodal: routine choice tasks (`wrong_tool` 6/6, `circular_delegation` 6/6) saturate; among recovery-action tasks, only `context_overflow`, `reasoning_loop`, and `stale_retrieval` reach 1/6 each, with zero detections on the remaining parsed recovery cases (item-level denominators in `real_llm_14fault.tsv`). LLM self-report and behavioral telemetry measure complementary surfaces; the 0.183 self-report rate quantifies a recovery-decision floor under which external typed telemetry would be required regardless of metamodel choice.

Detectable-significance threshold (not classical power).

At $n = 14$ under Holm-3 at family-wise $\alpha = 0.05$, the smallest discordance count for which all three pairs reach $p_{\text{adj}} \leq 0.05$ is $c \geq 7$ with $b = 0$ (raw $p = 0.0156$, Holm-3 0.0469, $\Delta_{\min} = 7/14 = 0.50$). At this floor, holding the baseline at its observed marginal 6/14 and asking what minimum DSM advantage is detectable, the cell counts are $(a, b, c, d) = (6, 0, 7, 1)$ giving $\text{DSM} = 13/14$ and $h_{\min} = 2 \arcsin \sqrt{13/14} - 2 \arcsin \sqrt{6/14} \approx 1.17$; the capability-envelope observation $c = 8$ has $h \approx 1.71$. Under survey-grounded coverage and conformant detectors, the realized cross-adapter difference between DSM and OpenInference extended attributes is $\Delta\text{TCR} = +0.024$ for DSM (cluster bootstrap 95% CI [0.000, +0.071]; cluster-level t test at $G - 1 = 5$ df two-sided $p = 0.36$; sign-flip permutation two-sided $p = 0.75$). The paper’s inferential apparatus is therefore designed to bound the detectable contrast under proper small- G treatment, not to establish equivalence; any realized effect smaller than the $h \approx 1.17$ floor is below the resolution of this design at $n = 14$.

5 Discussion

The conformance gap is a framework gap, not a metamodel gap.

The $4.5\times$ realized cross-adapter range (0.143 to 0.643) is the largest observed spread in the benchmark and comes from the source-survey-grounded framework profiles rather than from the DSM-vs-OpenInference extended contrast. `anthropic_sdk` sits at the bottom (no built-in tracing module; verified via full-text grep on `src/anthropic/`); `openai_sdk` sits at the top (typed `handoff` with `from_agent/to_agent` and typed `guardrail` with explicit `triggered`). The actionable open work is *per-framework instrumentation completeness*, not metamodel extension.

6 Threats to Validity

Construct validity.

TCR is binary per fault class on a rate-1 mocked-LLM setup; it measures metamodel-with-framework expressivity, not production recall. Some predicates use benchmark-oracle fields; the operational non-oracle subset is reported separately, and `telemetry_field_audit.tsv` classifies every predicate input. The vanilla OTel condition uses shared benchmark measurement attributes in addition to generic span kinds; it should not be read as a claim that OTel core standardizes agent diagnostics.

Internal validity (constructive bias).

The 14-fault benchmark and the nine DSM kinds were authored by us; there was no independent detector-bank or adapter-credit audit. Mitigations are limited but explicit: MAST mapping plus second coder ($\kappa = 0.83$, CI [0.58, 1.00]), freeze log, pinned-SHA source survey, third-party-only headline, per-kind ablation, and counter-experiments restricted to published attributes. FM-1.2 and FM-2.2 are unexercised and bound the claim.

External validity.

Adapter profiles are executable simulations grounded in source-survey evidence, not live package integrations; live SDK and CLI checks are sanity checks only. We claim no generalization beyond the surveyed versions, the 14-fault MAST-anchored taxonomy, mocked LLMs, and the three scoring rules. The headline FPR uses one no-fault workflow per (condition, adapter, model) cell; the supplemental six-workflow held-out check reduces threshold overfitting risk but remains synthetic and should not be read as an operational alert rate.

Conclusion validity.

The $n = 14$, $G = 6$ design resolves only large effects; the realized DSM-vs-OI contrast is statistically inconclusive, not equivalence evidence. We report Wilson CIs, descriptive paired statistics for the designed capability gap, exact adapter range, cluster bootstrap, cluster-level t , sign-flip, and GEE only as sensitivity.

7 Conclusion

Within this source-survey-grounded simulation and the surveyed SDKs, framework instrumentation defaults create the largest observed spread: realized DSM spans $4.5\times$, from 2/14 on Anthropic SDK to 9/14 on OpenAI Agents. Under published-attribute scoring, default-threshold DSM reaches (36/84, 4/72) versus OpenInference extended (34/84, 0/36); OTel GenAI extended reaches (38/84, 0/36), and an OpenInference EVALUATOR/PROMPT counterfactual reaches (40/84, 0/36). A supplemental six-workflow no-fault suite selects `max_retries=5` with held-out FPR 0/216 and moves DSM to (36/84, 0/72), but this remains a sensitivity operating point. Under small- G -appropriate inference the cross-adapter difference is statistically inconclusive (cluster bootstrap 95% CI on the per-adapter mean Δ is [0.000, +0.071]; cluster-level t test at $G - 1 = 5$ df two-sided $p = 0.36$; sign-flip $p = 0.75$). The 13-fault MAST-only sub-analysis matches direction and magnitude. The practical open work is per-framework instrumentation completeness, not metamodel extension.

8 Data Availability

The anonymized replication package contains the executable metamodel and invariants (`agentmm.ecore`, `agentmm.ocl`); source- survey and freeze evidence (`coverage_matrix.md`, `source_survey_evidence.tsv`, `live_sdk_validation.tsv`, `workload_design.md`, `mast_mapping*`); the executable adapter/fault harness, trace generator, schema, detectors, scorer, and three predicate banks (`adapter_harness.py`, `generate_trace_corpus.py`, `trace_schema.md`, `trace_detectors.py`, `score_traces.py`, `typed/permissive/extended_predicates.py`); audit tables (`predicates_lines.tsv`, `telemetry_field_audit.tsv`, `detector_fairness_checklist.tsv`, `kind_ablation.tsv`); analysis and sensitivity scripts (`analysis.py`, `compute_fpr.py`, `compute_roc.py`, `generate_no_fault_suite.py`, `threshold_holdout.py`, `eval_prompt_subexperiment.py`); cached trace/results data (`traces_full.jsonl`, `results_full.tsv`, 3,780 rows; `no_fault_suite_traces.jsonl`, `threshold_holdout.tsv`); cached live-LLM outputs (`real_llm_sanity.tsv`, `real_llm_14fault.tsv`); the Makefile; and `requirements.lock`.

The README’s two-line reproduction regenerates the trace corpus and all reported counts/statistics on one CPU in about three minutes; all bootstraps are seeded. The package is anonymized for review and will be deposited with a permanent DOI on Zenodo at camera-ready time.

Source-audit snapshot.

The full file/line ledger is `source_survey_evidence.tsv`; this compact snapshot records what is credited in the benchmark profile and what source evidence was found but not credited. A single judgment-sensitive advanced-kind flip changes the 84-cell aggregate by one cell for delegation/guardrail/reasoning/retrieval or two cells for memory ownership/corruption, so these rows are sensitivity targets but do not change the qualitative framework-gap conclusion.

Adapter	Credited evidence	Excluded or limited evidence
langchain	LLM, tool, retriever callbacks and <code>run_type</code> values.	Chain/generic metadata and reasoning-token usage lack DSM detector fields.
llamaindex	<code>AGENT_STEP</code> , LLM, <code>FUNCTION_CALL/TOOL</code> , and <code>RETRIEVE</code> callbacks.	Reasoning/thinking content is provider output, not a reasoning-step span; no memory/guardrail/planning/staleness fields.
autogen	GenAI create/invoke-agent spans, execute-tool spans, model-client calls, and handoff messages.	No detector fields for planning, reasoning-state repetition, guardrail result, memory ownership/corruption, or retrieval staleness.
crewai	LLM-call and tool-usage events in the tracing listener.	Optional agent, guardrail, memory, reasoning, and A2A events exist, but the default profile lacks the detector-specific fields used for DSM credit.
anthropic	Beta managed-agent model-request events; LLM evidence counted only under external instrumentation.	No built-in OTel tracer or detector fields for agent/tool/delegation, guardrail, memory, retrieval, planning, or reasoning.
openai	Agent, generation/response, function, handoff, and guardrail span data.	No planning, reasoning, retrieval-staleness, or memory ownership/corruption span-data classes.

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